SECTOR: FINANCE

Credit Risk Analysis

LendingClub Predictive Default Modeling | Newton School of Technology

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CONTEXT & PROBLEM STATEMENT

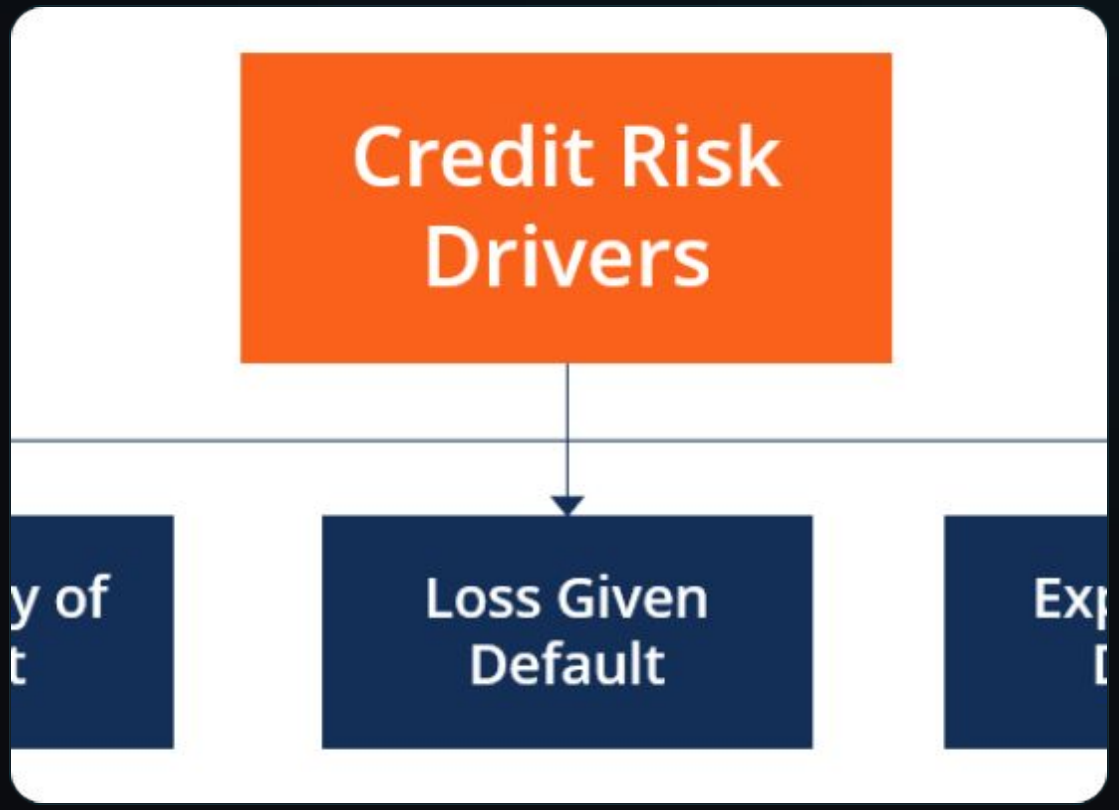
Business Challenge

LendingClub connects borrowers with investors but faces a major hurdle: **Principal Loss** from defaults. Accurate risk identification is critical for investor confidence.

Core Objective

Develop a predictive system to identify **high-risk borrowers** before capital deployment, aiming for 80% accuracy and 60% recall on defaults.

Stakeholder: Chief Risk Officer (CRO)



DATA ENGINEERING & ETL

Extraction

Processed ~396.03K raw records down to 391.95K cleaned samples from Kaggle LendingClub dataset.

Processing

Median imputation, log transformation, and 99th percentile capping for annual income outliers.

Feature Engineering

Developed 8 Stress Metrics: loan-to-income, installment-to-income, and composite credit stress scores.

Final Schema: 32 Features | Time Coverage: Historical Loan Outcomes

KPI & METRIC FRAMEWORK

14.8%
PORTFOLIO DEFAULT RATE

$750M
CHARGE-OFF LOSS EXPOSURE

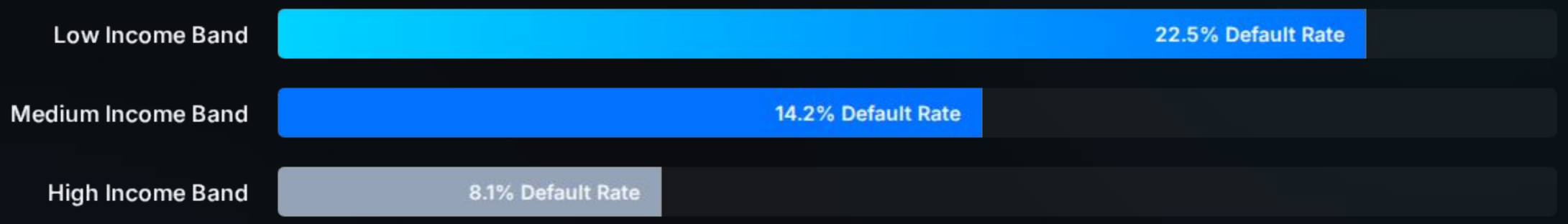
32
TOTAL RISK FEATURES

Avg DTI Ratio: Measures borrower leverage relative to monthly obligations.

Credit Stress Score: Composite metric of DTI and credit utilization.

Loan-Income Ratio: Determines affordability relative to debt size.

KEY EXPLORATORY INSIGHTS



Term Risk

60-month loans are nearly **twice as risky** as 36-month terms.

Utilization

Utilization > 70% shows extreme correlation with defaults.

DTI Spike

DTI > 20% results in **3x higher** likelihood of a "Charged Off" status.

STATISTICAL & MODEL ANALYSIS

Hypothesis Test	T-Statistic	P-Value	Result
Income Comparison (Paid vs Default)	51.827	0.000e+00	Highly Significant
DTI Comparison (Paid vs Default)	-20.911	7.875e-97	Highly Significant

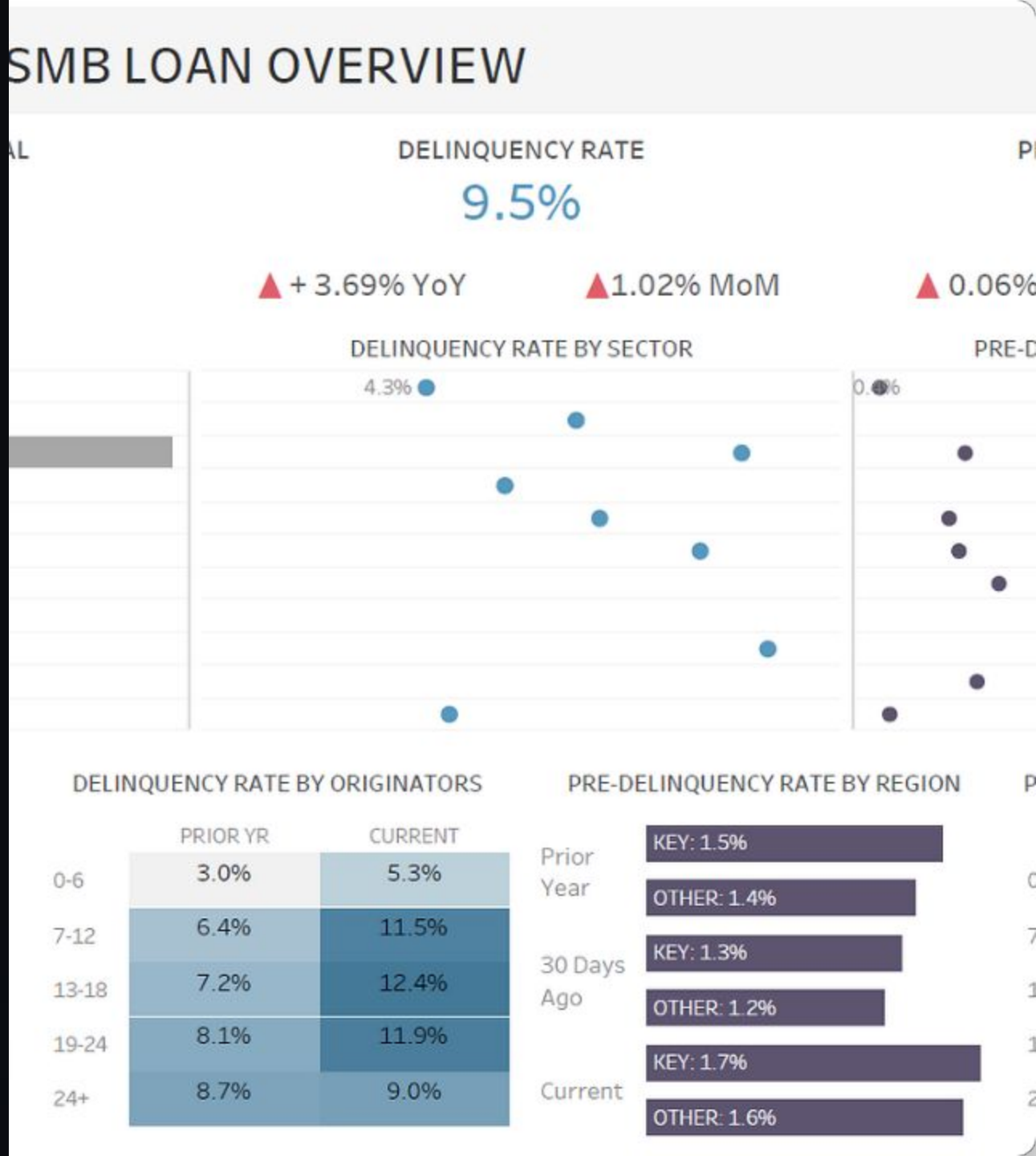
Logistic Regression
Accuracy: **80.35%**

Random Forest
Accuracy: **80.47%**

DASHBOARD OVERVIEW

Tableau Visualization
Interactive architecture supporting Executive monitoring and Operational drill-downs.

- **Filters:** Loan Grade, Year, Income Band
- **Views:** Geographic and Segment Analysis
- **Live:** Real-time portfolio exposure tracking



TOP DECISION INSIGHTS

Compounded Stress

Risk spikes exponentially when **both** High DTI and High Utilization coincide.

Tail Management

Top 1% of risk scores contribute to disproportionate losses; capping exposure is key.

Interest Rates

Interest rate is the strongest predictor of default, acting as the primary risk buffer.

BUSINESS RECOMMENDATIONS

- **Risk-Based Pricing:** Dynamically adjust interest rates for high-DTI and high-utilization segments to optimize returns.
- **Affordability Controls:** Implement a 25% "Hard Limit" for DTI ratios on all new loan applications.
- **Automated Underwriting:** Deploy the Random Forest model for real-time scoring and instant approval/rejection decisions.
- **Early Warning System:** Continuously monitor "Prime" borrowers who show a sudden utilization spike (>20% monthly increase).

EXPECTED BUSINESS IMPACT

15–20% Reduction: Projected default risk decrease using multi-factor filters.

Increased ROI: Via optimized risk-based pricing and automated efficiency.

Capital Stability: Reduction in portfolio volatility and charge-off losses.

"Moving beyond single-variable filters to composite stress scores stabilizes capital efficiency."

IMAGE SOURCES

https://cdn.corporatefinanceinstitute.com/assets/credit-risk-analysis.png

Source: corporatefinanceinstitute.com

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